

Estimating Covid-19 Dynamics and Clinical Detection Rate using an approximation to the discrete stochastic SEIR Model

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Introduction

The Covid-19 pandemic highlighted the challenge of imperfect detection in disease surveillance. Case reports undercount true infections due to asymptomatic cases and lack of access to testing. As a result, typically only the sickest and those with access to care got tested for and diagnosed with Covid-19 early in the pandemic. Various approaches have been developed in order to estimate the Covid-19 clinical detection rate including seroprevalence studies (Taylor et al. 2023; Samore et al. 2021), multiplier models (Reese et al. 2021), back-calculation from deaths or hospitalizations, and most commonly, dynamic transmission models or compartment models which jointly estimate disease dynamics and clinical detection rate (Deo and Grover 2021; Chitwood et al. 2022). These dynamic transmission models are typically ODE-based models, which correspond to large-population limits of more realistic stochastic transmission models with discrete state spaces (CITE needed).

Despite the increased fidelity of stochastic transmission models, these models present computational challenges due to the fact that their discrete state space is only partially observed, necessitating marginalization over a high dimensional discrete space. Several approaches exist to overcome this challenge.

Recent innovations include linear noise approximations (Fintzi, Wakefield, and Minin 2022), particle filtering and iterated filtering (Li et al. 2024), Dirichlet distributed compartment models (Zhou et al. 2020), and exact marginalization using Laplace Transformations (Ho, Crawford, and Suchard 2018). While all of the aforementioned work addresses the challenge, gaps remain.

As Bayesians, we would like to build a rich stochastic model that incorporates geographic heterogeneity, under-reporting biases, and time-varying transmission dynamics, as well as efficiently sampling from the posterior for parameters of interest. This still remains a hurdle for these prior works. Our approach is to approximate the discrete dynamics with a continuous process, akin to the linear noise approximations but in discrete time, which obviates some of

the computational issues with having to repeatedly solve an ODE. Because our approach is developed in Stan, we can apply principles of Bayesian model building, as outlined in [Bayesian workflow paper], and formulate a model that reaches a good tradeoff between verisimilitude and tractability.

Covid-19’s incubation period, during which individuals are infected but not infectious, cannot be captured by the standard SIR model (Susceptible–Infected–Recovered). As a result SEIR models (Susceptible–Exposed–Infected–Recovered), which allow for a pre-infectious incubation period, align more closely with Covid-19’s biology. As a result, these compartment models became the main models for Covid-19 projections during the pandemic (Yang et al. 2020). Although the SEIR model leads to more accurate timing of outbreak progression by incorporating the pre-infectious latency period, it adds to the number of dynamics parameters that must be estimated without additional data supporting those estimations. Also, these models do not directly account for imperfect detection. Various models formulations, both stochastic and deterministic, have been devised to include compartments for both ascertained and un-ascertained infecteds, allowing for the estimation of under-ascertainment and differing rates of infectiousness of these compartments (Hao et al. 2020; Chitwood et al. 2022).

In this study, we devise a model, implemented using the No-U-Turn Sampler (NUTS), a variant of Hamiltonian Monte Carlo (HMC), that uses simple case counts reported at the health-district level in Utah to estimate the proportion of COVID-19 infections that are detected, i.e., the clinical detection rate. Our model introduces three novel methodological advantages:

- (1) model estimation allows for different start times of infections across health-districts,
- (2) allowing an estimable time-evolving and country-specific transmission rate using a hierarchical AR(1) process , and
- (3) a continuous approximation that enables gradient-based Bayesian inference of dynamics in a discrete-time stochastic SEIR with beta-binomial transitions.

Methods

We fit a stochastic model on weekly new case counts across 11 health districts in Utah based on the SEIR (Susceptible–Exposed–Infected–Removed) compartmental model used to simulate Covid-19 infections. Given the complexity of Covid-19 case report data due to shifting policy and safety precautions, our approach required flexibility in the dynamics between compartments over time and in space. Specifically, we fit a model which allowed the force of infection to change over time and for it and the rate of movement between E and I compartments and I and R compartments to differ between districts. Finally, our model does not assume that the dynamics in the SEIR compartmental model start at the same time as cases reported in Utah at the district level do not start in the same week.

Model Formulation

We assume incident cases of Covid-19 follow a stochastic SEIR process with underreporting. Let N_i be the total population at risk at time 1 for district i . Let S_{it} , E_{it} , I_{it} denote susceptible, exposed, and infected populations. Let SE_{it} , EI_{it} , IR_{it} be transitions. Each district has a first reporting time f_i .

$$\begin{aligned} I_{if_i} &= c_{f_i}, \\ EI_{if_i} &= I_{if_i}, \\ S_{if_i} &= N_i - EI_{if_i}. \end{aligned}$$

For $(t > f_i)$:

$$\begin{aligned} SE_{it} &\sim \text{Binomial}\left(S_{it-1}, 1 - \exp\left(-\frac{\beta_{it}I_{it-1}}{N}\right)\right), \\ EI_{it} &\sim \text{BetaBinomial}(E_{it-1}, 1 - \exp(-\eta_i), \rho_{EI}), \\ IR_{it} &\sim \text{BetaBinomial}(I_{it-1}, 1 - \exp(-\gamma_i), \rho_{IR}), \\ S_{it} &= S_{it-1} - SE_{it}, \\ E_{it} &= \begin{cases} SE_{it}, & t = f_i + 1, \\ E_{it-1} + SE_{it} - EI_{it}, & t > f_i + 1 \end{cases} \\ I_{it} &= I_{it-1} + EI_{it} - IR_{it}, \\ i_{it} &\sim \text{Binomial}(EI_{it}, p). \end{aligned}$$

We assume imperfect detection with constant detection probability (p). A supplemental model uses a beta-binomial observation process.

Transmission follows:

$$\begin{aligned} \log \beta_{it} &\sim \text{Normal}(B_{it}, \tau^2) \\ B_{it} &= \phi B_{i,t-1} + \sigma \epsilon_{it}, \quad t > 1 \end{aligned}$$

where $\epsilon_{it} \sim \text{Normal}(0, 1)$.

Hamiltonian Monte Carlo Approximation

Since NUTS requires gradients, discrete latent variables are approximated by continuous processes.

First, binomial detection is approximated by Gaussian distributions (B. Brintz, Fuentes, and Madsen 2018; B. J. Brintz, Madsen, and Fuentes 2023).

Second, transitions are modeled on the proportion scale by dividing by N_i . Let lowercase letters denote proportions.

We approximate $se_{it}, ei_{it}, ir_{it}$ with scaled inverse-logits of normal random variables matched on mean and variance (see Appendix).

Results

We fit the model to 30 weeks of data from 11 health districts in Utah. The overall posterior mean clinical detection fraction was 0.16 (90% CrI: 0.14–0.17), indicating substantive under-ascertainment of infections during the study period. Weekly posterior estimates show the effect of under-ascertainment on observed case counts (Figure 1). The hierarchical AR(1) structure on log-beta captured changes in transmission intensity over time with an estimated mean (B_{it}) of 0.11 (90% CrI: -0.05-0.27) and standard deviation (σ) of 0.54 (90% CrI: 0.43-0.66). In addition to AR(1) correlated ($\phi=0.81$ (90% CrI: 0.72-0.88)) transmission, we observed heterogeneity across health districts in timing of epidemic growth, driven by differences in exposure stage to infection stage with estimates between 0.26 and 0.87 and recovery rate between 0.2 and 0.68 between districts with beta-binomial overdispersion parameters of 0.01 (90% CrI: 0-0.01) and 0.77 (90% CrI: 0.43-0.98), respectively.

Model diagnostics were favorable: no divergent transitions were observed and E-BFMI was within 0.1 of 1 for all chains. Together these results indicate a consistent signal of under-detection across Utah with temporal and spatial variation that our hierarchical model can capture.

Simulation Study

We used simulated data to assess model performance in recovering true parameters and states. Similar to the Utah health district data, we simulated 30 weeks of data for 11 districts with known parameters. In 200 simulations of observed case counts per setting, we varied the detection probability ($p = 0.15, 0.50, 0.85$), used true health district population data, and set true parameters to match posterior draws from the Utah data model fit. Across simulated detection probabilities, we found the model recovered parameters well with low average error, high coverage, and reasonable credible interval widths across all detection probabilities. As expected, higher detection probabilities led to improved parameter recovery (Table 1).

Table 1: Simulation results for key parameters across detection probabilities.

Detection Probability	Average Error (%)	Coverage	90% CrI Width
Low (0.15)	0.002 (1.3%)	0.927	0.008

Table 1: Simulation results for key parameters across detection probabilities.

Detection Probability	Average Error (%)	Coverage	90% CrI Width
Medium (0.50)	0.006 (1.2%)	0.876	0.026
High (0.85)	0.01 (1.2%)	0.778	0.041

While our case study model fit had no divergences, the simulation study had some divergences across detection probabilities (Table 2). On average, there were low divergence rates across all detection probabilities, with higher rates at higher detection probabilities.

Table 2: Summary of divergences across 200 simulations for each detection probability.

Detection Probability	Mean (SD)	Percentile				
		2.5th	25th	50th	75th	97.5th
Low (0.15)	0.031 (0.043)	0	0.004	0.017	0.041	0.166
Medium (0.50)	0.029 (0.046)	0	0.004	0.012	0.036	0.143
High (0.85)	0.084 (0.135)	0	0.006	0.020	0.081	0.506

Discussion

We developed a stochastic SEIR model using a continuous approximation to discrete transitions to estimate the clinical detection rate of Covid-19 in Utah health districts. Our model estimated a clinical detection rate of 16% (90% CrI: 14–17%), indicating substantial underascertainment during the study period. The hierarchical AR(1) structure on transmission rates effectively captured temporal changes and spatial heterogeneity across districts.

Appendix: Continuous Approximation of Discrete Transitions

Appendix: Inverse Logit calculation

$$\begin{aligned}
\mathbb{E} \left[\frac{SE_t}{N} \right] &= s_{t-1}(1 - \exp(-\beta i_{t-1})), & \text{Var} \left(\frac{SE_t}{N} \right) &= \frac{s_{t-1}}{N} \exp(-\beta i_{t-1})(1 - \exp(-\beta i_{t-1})) \\
\mathbb{E} \left[\frac{EI_t}{N} \right] &= e_{t-1}(1 - \exp(-\eta)), & \text{Var} \left(\frac{EI_t}{N} \right) &= \frac{e_{t-1}}{N} \exp(-\eta)(1 - \exp(-\eta)) \\
\mathbb{E} \left[\frac{IR_t}{N} \right] &= i_{t-1}(1 - e^{-\gamma}), & \text{Var} \left(\frac{IR_t}{N} \right) &= \frac{i_{t-1}}{N} e^{-\gamma}(1 - e^{-\gamma})
\end{aligned}$$

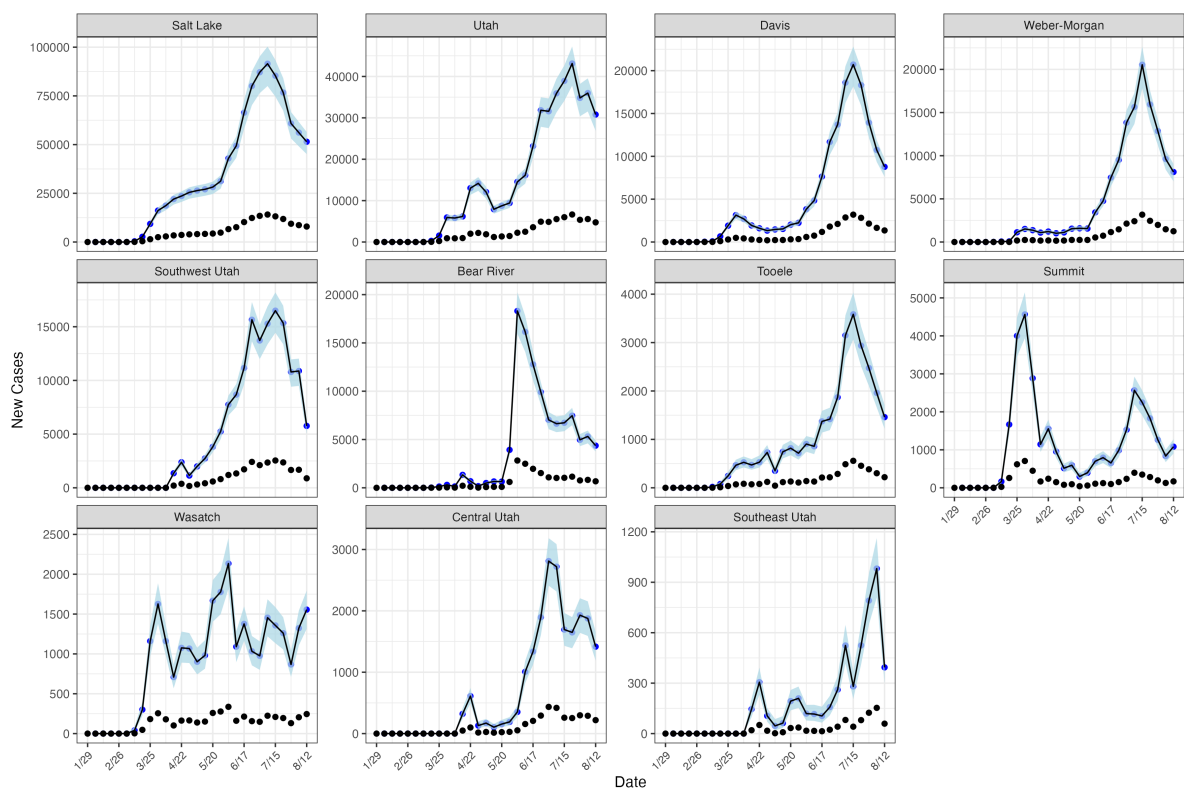


Figure 1: Weekly observed cases reported (black) and model estimated cases (blue) for Utah health district.

Let se_t , ei_t , and ir_t be random variables with bounds $[0, s_{t-1}]$, $[0, e_{t-1}]$, and $[0, i_{t-1}]$, respectively. We need to find u_t , v_t , and w_t distributed on $[0, 1]$ such that $se_t = u_t s_{t-1}$, $ei_t = v_t e_{t-1}$, and $ir_t = i_{t-1} w_t$ with the same mean and variance.

$$\begin{aligned}
\mathbb{E}[s_{t-1}u_t] &= s_{t-1}\mathbb{E}[u_t] = s_{t-1}(1 - \exp(-\beta i_{t-1})) \\
\text{Var}(s_{t-1}u_t) &= s_{t-1}^2 \text{Var}(u_t) = \frac{s_{t-1}^2}{N} \exp(-\beta i_{t-1})(1 - \exp(-\beta i_{t-1})) \\
\mathbb{E}[e_{t-1}v_t] &= e_{t-1}\mathbb{E}[v_t] = e_{t-1}(1 - \exp(-\eta)) \\
\text{Var}(e_{t-1}v_t) &= e_{t-1}^2 \text{Var}(v_t) = \frac{e_{t-1}^2}{N} \exp(-\eta)(1 - \exp(-\eta)) \\
\mathbb{E}[i_{t-1}w_t] &= i_{t-1}\mathbb{E}[w_t] = i_{t-1}(1 - e^{-\gamma}) \\
\text{Var}(i_{t-1}w_t) &= i_{t-1}^2 \text{Var}(w_t) = \frac{i_{t-1}^2}{N} e^{-\gamma}(1 - e^{-\gamma})
\end{aligned}$$

Solving for $\mathbb{E}[u_t]$, $\text{Var}(u_t)$, $\mathbb{E}[v_t]$, $\text{Var}(v_t)$, $\mathbb{E}[w_t]$, and $\text{Var}(w_t)$ implies

$$\begin{aligned}
\mathbb{E}[u_t] &= (1 - \exp(-\beta i_{t-1})) \\
\text{Var}(u_t) &= \frac{1}{s_{t-1}N} \exp(-\beta i_{t-1})(1 - \exp(-\beta i_{t-1})) \\
\mathbb{E}[v_t] &= (1 - \exp(-\eta)) \\
\text{Var}(v_t) &= \frac{1}{e_{t-1}N} \exp(-\eta)(1 - \exp(-\eta)) \\
\mathbb{E}[w_t] &= (1 - e^{-\gamma}) \\
\text{Var}(w_t) &= \frac{1}{i_{t-1}N} e^{-\gamma}(1 - e^{-\gamma})
\end{aligned}$$

What we could do is to take $\tilde{u}_t, \tilde{v}_t, \tilde{w}_t \sim \text{Normal}$ and let $u_t = \text{inv_logit}(\tilde{u}_t)$, $v_t = \text{inv_logit}(\tilde{v}_t)$, and $w_t = \text{inv_logit}(\tilde{w}_t)$.

The downside of this approach is that the expectation of the inverse logit of a normal does not have an analytic form, but we could approximate with the Delta method. Let

$$\tilde{u}_t \sim \text{Normal}(\mu_u, \sigma_u^2), \tilde{v}_t \sim \text{Normal}(\mu_v, \sigma_v^2), \tilde{w}_t \sim \text{Normal}(\mu_w, \sigma_w^2).$$

We need to find $\mu_u, \sigma_u^2, \mu_v, \sigma_v^2, \mu_w, \sigma_w^2$ such that

$$\begin{aligned}
\mathbb{E}[\text{expit}(\tilde{u}_t)] &\approx (1 - \exp(-\beta i_{t-1})), \quad \text{Var}(\text{expit}(\tilde{u}_t)) \approx \frac{1}{s_{t-1}N} \exp(-\beta i_{t-1})(1 - \exp(-\beta i_{t-1})) \\
\mathbb{E}[\text{expit}(\tilde{v}_t)] &\approx (1 - \exp(-\eta)), \quad \text{Var}(\text{expit}(\tilde{v}_t)) \approx \frac{1}{e_{t-1}N} \exp(-\eta)(1 - \exp(-\eta)) \\
\mathbb{E}[\text{expit}(\tilde{w}_t)] &\approx (1 - e^{-\gamma}), \quad \text{Var}(\text{expit}(\tilde{w}_t)) \approx \frac{1}{i_{t-1}N} e^{-\gamma}(1 - e^{-\gamma})
\end{aligned}$$

Using the delta method gives that

$$\begin{aligned}
\mathbb{E}[\text{expit}(\tilde{u}_t)] &\approx \text{expit}(\mu_u), \quad \text{Var}(\text{expit}(\tilde{u}_t)) \approx \sigma_u^2 (\text{expit}(\mu_u)')^2 \\
\mathbb{E}[\text{expit}(\tilde{v}_t)] &\approx \text{expit}(\mu_v), \quad \text{Var}(\text{expit}(\tilde{v}_t)) \approx \sigma_v^2 (\text{expit}(\mu_v)')^2 \\
\mathbb{E}[\text{expit}(\tilde{w}_t)] &\approx \text{expit}(\mu_w), \quad \text{Var}(\text{expit}(\tilde{w}_t)) \approx \sigma_w^2 (\text{expit}(\mu_w)')^2
\end{aligned}$$

This leads to the following parameters:

$$\begin{aligned}
\mu_u &= \text{logit}(1 - \exp(-\beta i_{t-1})) \\
\sigma_u^2 &= \frac{1}{s_{t-1}N} \frac{\exp(-\beta i_{t-1})(1 - \exp(-\beta i_{t-1}))}{((1 - \text{expit}(\mu_u))\text{expit}(\mu_u))^2} \\
&= \frac{1}{\exp(-\beta i_{t-1})(1 - \exp(-\beta i_{t-1}))s_{t-1}N} \\
\mu_v &= \text{logit}(1 - \exp(-\eta)) \\
\sigma_v^2 &= \frac{1}{e_{t-1}N} \frac{\exp(-\eta)(1 - \exp(-\eta))}{((1 - \text{expit}(\mu_v))\text{expit}(\mu_v))^2} \\
&= \frac{1}{\exp(-\eta)(1 - \exp(-\eta))e_{t-1}N} \\
\mu_w &= \text{logit}(1 - e^{-\gamma}) \\
\sigma_w^2 &= \frac{1}{i_{t-1}N} \frac{e^{-\gamma}(1 - e^{-\gamma})}{((1 - \text{expit}(\mu_w))\text{expit}(\mu_w))^2} \\
&= \frac{1}{e^{-\gamma}(1 - e^{-\gamma})i_{t-1}N}
\end{aligned}$$

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